

THREE CONJECTURES ON ALTERNATING PLANE GRAPHS

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ABSTRACT. In an alternating plane graph, adjacent vertices have different degrees and adjacent faces have different sizes. Four problems on these graphs were left open by Althöfer, Haugland, Scherer, Schneider and Van Cleemput. We settle two and advance a third.

No alternating plane graph has exactly two degrees or exactly two face sizes (Conjecture 10.1). One per-edge inequality, applied twice, proves this, and the result survives when bridges are allowed.

Conjecture 10.2, that $(3, 4, 5)$ -alternating plane graphs exist for every $n \geq 20$, is settled. We give verified certificates at the 26 orders that were open, and a periodic capping lemma reaching order 48 and every $n \geq 50$. Together these close every $n \geq 46$ independently of the original paper.

For Conjecture 10.3 we give 3-connected witnesses at 54 orders. Its infinite tail is open, and so is the fourth problem.

1. INTRODUCTION

An alternating plane graph is a plane graph in which neighbours differ: adjacent vertices have different degrees, and adjacent faces have different sizes. Althöfer, Haugland, Scherer, Schneider and Van Cleemput [1] introduced these graphs and closed their paper with four open problems. We settle the first two and advance the third.

Conjecture 10.1 says there are no $2, Y$ -alternating and no $X, 2$ -alternating plane graphs, where an X, Y -alternating plane graph has exactly X distinct degrees and exactly Y distinct face sizes. Althöfer et al. showed that a $2, Y$ -alternating plane graph has degrees 3 and 4, or 3 and 5, with at least 25 or 56 vertices respectively. For weak alternating plane graphs, which allow degree 2, they bounded the number of faces by summing $1/r + 1/s$ over edges between an r -face and an s -face.

That face count never uses degree 2, and it is most of the proof. Add the same count over the two ends of each edge. By Euler's formula the four reciprocals, summed over all edges, give $e + 2$, so some edge carries more than 1. Alternation alone allows $\frac{1}{3} + \frac{1}{4}$ on each side, which is $\frac{7}{6}$ and too much. But two values on one side make the graph bipartite there, so the values on the other side are even. Distinct even values are at least 4 and 6, and $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$. Now $\frac{7}{12} + \frac{5}{12} = 1$ at every edge, a contradiction (Theorem 4.1). In §5 we allow bridges. A bridge sits on a face of size at least 8, and that pays for the few edges that can exceed 1.

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Conjecture 10.2 asks for a $(3, 4, 5)$ -alternating plane graph on every $n \geq 20$. Twenty-six orders were open, and we give certificates for all of them. They are capped pieces of one periodic strip, and the capping lemma (Theorem 6.1) shows that inserting further periods keeps the graph alternating, because every face lies inside five consecutive periods.

Conjecture 10.3 asks for 3-connected alternating plane graphs. We give witnesses at 54 orders (Theorem 7.1). The spliced graphs above order 56 are not known to be 3-connected, and $(3, 4, 5)$ -alternating plane graphs need not be (Proposition 7.2). The asymptotic degree distribution stays open (§8).

The proofs in §§3–5 use no computer. The certificates run to 110 vertices. We give them as rotation systems and recompute every property from these. The capping lemma rests on finite checks of a single certificate, listed in Remark 6.2.

2. PRELIMINARIES

Definition 2.1 ([1], Definition 2.1). An *alternating plane graph* is a simple connected plane graph in which

- (a) every vertex has degree at least 3;
- (b) every face, the unbounded one included, has size at least 3;
- (c) adjacent vertices have different degrees;
- (d) adjacent faces have different sizes.

Definition 2.2 ([1], Definitions 3.1 and 3.4). A $(3, 4, 5)$ -alternating plane graph is one in which every vertex degree and every face size lies in $\{3, 4, 5\}$. An X, Y -alternating plane graph is one with exactly X distinct vertex degrees and exactly Y distinct face sizes.

Convention 2.3 (face size). The *size* of a face is the length of its boundary walk counted with multiplicity [3, §3.3], that is, the number of edge-side incidences. A bridge lies on one face from both sides and contributes 2 to its size.

In a bipartite plane graph every closed walk is even, so every face has even size under this convention. Counting distinct boundary vertices would lose that. This is also the count behind $\sum_s s f_s = 2e$ in (3.1) and (9.2) of [1].

Convention 2.4 (bridges). Say that a plane graph satisfies (C2) if no edge has the same face on both sides, equivalently if it has no bridge. In (d), the faces on the two sides of an edge must have different sizes, so a bridge breaks (d) and alternating plane graphs satisfy (C2) [1]. Under the *weak reading*, (d) constrains only pairs of distinct faces and bridges are permitted.

Section 4 uses (C2), and §5 proves the same result under the weak reading.

Theorem 2.5 ([1], Theorem 3.2). *In a $(3, 4, 5)$ -alternating plane graph, $v_3 = f_3$, $v_4 = f_4$ and $v_5 = f_5$, where v_i and f_i count vertices of degree i and faces of size i . Consequently $v_5 = v_3 - 4$, $E = 2V - 2$ and $F = V$.*

2.1. Rotation systems. A plane map is encoded by its *rotation system* [3]: for each vertex, its neighbours in clockwise order. It determines the map up to homeomorphism. Write α for the involution exchanging the two darts of an edge and σ for the rotation. The faces are the orbits of $\varphi = \sigma^{-1}\alpha$, and

the size of a face is the length of its orbit. An edge is a bridge if and only if its two darts lie in the same orbit.

3. THE PER-EDGE IDENTITY

Lemma 3.1. *Let G be a finite plane graph with no isolated vertex, with v vertices, e edges and f faces. For an edge a with ends u, w and face occurrences $F^-(a), F^+(a)$ on its two sides, put*

$$c(a) = \frac{1}{\deg u} + \frac{1}{\deg w} + \frac{1}{|F^-(a)|} + \frac{1}{|F^+(a)|}.$$

Then $\sum_a c(a) = v + f$. If G is connected, Euler's formula gives

$$\sum_a c(a) = e + 2, \quad \text{equivalently} \quad \sum_a (c(a) - 1) = 2.$$

Proof. Each vertex hands $1/\deg$ to each edge end at it, and each face hands $1/|F|$ to each edge side on its boundary walk (Convention 2.3). Each pays out exactly 1, so the edges receive $v + f$. $\square$

Neither count needs facial walks to be simple. A bridge has $F^-(a) = F^+(a)$ and receives $2/|F|$ from that face, so the lemma holds with bridges too, and §5 uses it that way.

We write $x(a) = c(a) - 1$ for the *excess* of an edge. So Lemma 3.1 reads $\sum_a x(a) = 2$, and in a connected plane graph some edge has positive excess.

4. CONJECTURE 10.1

Theorem 4.1. *There is no 2, Y -alternating plane graph and no $X, 2$ -alternating plane graph.*

The two halves exchange vertices and faces, but no dual graph is used, so neither needs the 3-edge-connectivity of [1, Lemma 2.2]. We prove both.

4.1. No 2, Y -alternating plane graph. Suppose G is one, with degrees $d_1 < d_2$. By (c) every edge joins a d_1 -vertex to a d_2 -vertex. So G is bipartite, and since $d_1 \geq 3$ the vertex side of $c(a)$ is at most

$$\frac{1}{d_1} + \frac{1}{d_2} \leq \frac{1}{3} + \frac{1}{4} = \frac{7}{12}.$$

Every closed walk in G is even, so every face has even size, at least 4. By (C2) the two sides of an edge are distinct faces, and by (d) their sizes differ. So one is at least 4 and the other at least 6, and the face side is at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$. Hence $c(a) \leq 1$ for every edge, against Lemma 3.1. $\square$

Remark 4.2. This is (9.5) of [1] with $1/2$ replaced by $1/d_1 \leq 1/3$. In §9.1 Althöfer et al. run the same chain on the weak class with degrees 2 and k . With $\frac{1}{2}$ kept, it excludes exactly $k \geq 12$, which is their (9.10). It does not exclude $k \leq 10$, where they exhibit weak 2, k -alternating plane graphs, and it leaves $k = 11$ open, as they do without their Lemma 9.4.

4.2. No $X, 2$ -alternating plane graph. Suppose G is one, with face sizes $s_1 < s_2$. By (C2) and (d) every edge separates an s_1 -face from an s_2 -face, so the face side is at most $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$.

Every degree is even. Fix a vertex u of degree d , let its edges in rotation order be $a_1, \dots, a_d$, and let F_i be the face in the corner between a_i and a_{i+1} , indices modulo d . The two sides of a_i are F_{i-1} and F_i , distinct faces by (C2), so $|F_{i-1}| \neq |F_i|$. With only two sizes the cyclic word $|F_1|, \dots, |F_d|$ alternates, so d is even.

Degrees are at least 3 and even, hence at least 4, and adjacent degrees differ. So the vertex side is at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$, and again $c(a) \leq 1$ for every edge. $\square$

5. REMOVING THE DEPENDENCE ON (C2)

Throughout this section we take the weak reading of Convention 2.4. A bridge breaks both halves of the argument. Its two sides are one face, and its ends need not have even degree. What rescues it is that a bridge sits on a face of size at least 8. So every bridge has excess at most $-\frac{1}{6}$, while a positive edge has excess at most $\frac{1}{24}$ and has a bridge at its degree-3 end. Each bridge takes at most two positive edges, and the bridges win.

Lemma 5.1 (parity). *Let G be a plane graph whose faces admit a colouring by two colours that is proper on every pair of distinct adjacent faces. Then for every vertex v ,*

$$\deg v - |\{\text{bridges at } v\}| \text{ is even.}$$

Proof. Take the corner faces $F_1, \dots, F_d$ at v as in §4.2. The occurrences F_{i-1} and F_i are the same face exactly when a_i is a bridge. So going once round v , the colour changes across each non-bridge edge and stays the same across each bridge. We return to the first colour, so the number of changes is even. $\square$

With no bridges, every degree is even. In an $X, 2$ -alternating plane graph the face size gives the colouring.

Corollary 5.2. *In an $X, 2$ -alternating plane graph under the weak reading, a vertex lying on no bridge has even degree, hence degree at least 4.*

Theorem 5.3. *Under the weak reading there is no $X, 2$ -alternating plane graph. In particular no such graph has a bridge.*

Proof. Let G be one, with face sizes $s_1 < s_2$, and let B be its number of bridges. Suppose first $B \geq 1$.

Step 0: $s_2 \geq 8$. Let $a = uw$ be a bridge, and let $A \ni u$ and $C \ni w$ be the components of $G - a$. Then $\deg_A u \geq 2$ and $\deg_C w \geq 2$. Let W_A be the boundary walk of A through the corner left by deleting a . It cannot have length 1 (a loop) or 2 (parallel edges, or $A = K_2$, but $\deg_A u \geq 2$). So $|W_A| \geq 3$, and likewise $|W_C| \geq 3$. The face F of G at a runs along W_A , across a , along W_C and back across a , so

$$|F| = |W_A| + |W_C| + 2 \geq 3 + 3 + 2 = 8.$$

As $|F| \in \{s_1, s_2\}$, we get $s_2 \geq 8$.

Step 1: only an edge at a vertex of degree 3 can have positive excess. We have $s_1 \geq 3$, $s_2 \geq 8$, and the two ends of an edge have distinct degrees. Three cases:

- a a bridge: both sides carry F , so $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6}$ and $x(a) \leq -\frac{1}{6}$;
- a not a bridge, both ends of degree at least 4: $c(a) \leq \frac{1}{4} + \frac{1}{5} + \frac{1}{3} + \frac{1}{8} = \frac{109}{120}$ and $x(a) \leq -\frac{11}{120}$;
- a not a bridge with an end of degree 3: $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{3} + \frac{1}{8} = \frac{25}{24}$ and $x(a) \leq \frac{1}{24}$.

Only the third class can be positive.

Step 2: each bridge carries at most two positive edges. This is the hard step. A positive edge is a 3–4 edge: its face side is at most $\frac{1}{3} + \frac{1}{8} = \frac{11}{24}$, so its vertex side exceeds $\frac{13}{24}$, and $\frac{1}{3} + \frac{1}{5} = \frac{12.8}{24}$ already falls short.

Let a be a positive edge and u its end of degree 3. By Lemma 5.1 the number of bridges at u is odd. It is not 3, since a is not a bridge. So u lies on exactly one bridge $b(u)$, and we charge a to $b(u)$.

Now fix a bridge b . Every positive edge charged to b has its degree-3 end on b . By (c) at most one end of b has degree 3, and at that end only two edges other than b remain. Writing P for the number of positive edges, $P \leq 2B$.

Step 3. Every other edge has $x(a) \leq 0$. So by Steps 1 and 2 and Lemma 3.1,

$$2 = \sum_a x(a) \leq \frac{P}{24} - \frac{B}{6} \leq \frac{2B}{24} - \frac{B}{6} = -\frac{B}{12} < 0,$$

a contradiction.

Finally suppose $B = 0$. By Corollary 5.2 every degree is even and at least 4, so the vertex side of every edge is at most $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ and the face side at most $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$. Hence $x(a) \leq 0$ for every edge, against $\sum_a x(a) = 2$. $\square$

Remark 5.4. Step 2 has room to spare. Without the degree condition at the ends of a bridge we get only $P \leq 4B$. The sum in Step 3 is then at most 0, which still contradicts 2. At $P \leq 5B$ the argument fails. Step 0 has room too: with $s_2 \geq 7$ a bridge and its two positive edges total at most $-\frac{1}{84}$, but with $s_2 \geq 6$ the bound is $+\frac{1}{12}$.

Proposition 5.5. *Under the weak reading there is no 2, Y -alternating plane graph either.*

Proof. Step 0 above uses only minimum degree 3 and simplicity, so a bridge's face has size at least 8. As in §4 the graph is bipartite, so every closed walk is even and every face has even size at least 4. A bridge has $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{2}{8} = \frac{5}{6}$. A non-bridge edge has two distinct faces of distinct even sizes on its sides, so $c(a) \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} = 1$. So $x(a) \leq 0$ for every edge, against $\sum_a x(a) = 2$. $\square$

Remark 5.6. Weak-reading alternating plane graphs with bridges exist. Join three copies of a (3, 4, 5)-alternating plane graph by two bridges drawn in the exterior, at a degree-5 vertex of each. The joined degrees become 7, 6, 6, their old neighbours keep degree 3 or 4, and the merged exterior face

has size at least 13, so it differs from every interior face. This graph has neither exactly two degrees nor exactly two face sizes.

6. THE PERIODIC CAPPING LEMMA

Conjecture 10.2 asks for a $(3, 4, 5)$ -alternating plane graph on n vertices for every $n \geq 20$. Twenty-six orders were open. We supply certificates for all of them (§9), and prove that a single certificate generates an infinite family.

6.1. The cover. The certificates are capped unrollings of a periodic quotient of the torus. The relevant structure is a bi-infinite *strip*, built from identical *copies*, each contributing three vertices and six edges, together with two *caps* that close it off at the ends. Write TARGET_n for a certificate of order n . Its alignment against the cover exhibits a maximal run of copies whose rotations are translates of one another; we call this run the *deep block* D .

Theorem 6.1 (periodic capping lemma). *Fix an order n whose alignment against the cover has a deep block D of at least five copies, and let cut be a copy at distance at least two from each end of D . Then for every integer $d \geq -(|D| - 1)$ the rotation system $\text{splice}(n, d)$, obtained from TARGET_n by inserting d further translates of the copy cut (or deleting $-d$ of them when $d < 0$), is a $(3, 4, 5)$ -alternating plane graph on $n + 3d$ vertices.*

Proof. Write $S = \text{splice}(n, d)$. Every vertex of S gets its rotation in one of four ways. Cap vertices keep theirs, with strip neighbours above the cut shifted by d . Strip copies at or below the cut keep theirs. A strip copy c above cut $+d$ takes the rotation of copy $c - d$. The d fresh copies take the rotation of cut, translated.

The argument is local. A facial walk is traced by $\varphi = \sigma^{-1}\alpha$, which reads only the edge partner and the rotation predecessor at each vertex it visits.

We bound face spans in TARGET_n , since the faces of S are what is in question. In TARGET_n every face has at most five darts, and the cover's largest edge offset is two. A closed sequence of at most five steps, each of size at most two, has span at most four: reaching level 5 needs three ascents, leaving two steps to descend five. So *every face of TARGET_n lies inside five consecutive copies*. Both inputs are finite checks on the one certificate (Remark 6.2). The bound four is not sharp; the largest span attained, over every spliceable order at $d = 1, 2, 3, 5$, is two.

The fresh copies lie in $(\text{cut}, \text{cut} + d]$ and are translates of cut. Since cut is at distance at least two from each end of D , the copies $\text{cut} - 2, \dots, \text{cut} + 2$ lie in D and are mutual translates. Hence every copy in $[\text{cut} - 2, \text{cut} + d + 2]$ is a translate of cut, and every five-copy window meeting the fresh block is a translate of $[\text{cut} - 2, \text{cut} + 2]$. Away from that range S agrees with TARGET_n .

Now trace φ in S from any dart. Its counterpart in TARGET_n closes within five steps inside a five-copy window, on which S is a translate of TARGET_n . So the walk in S closes at the same length, and *every face of S is a translated face of TARGET_n* .

Each condition of Definition 2.1 is local, so S inherits it: face sizes and degrees lie in $\{3, 4, 5\}$, adjacent vertices have unequal degrees and adjacent

faces unequal sizes, and facial walks are simple. A loop or parallel edge would sit inside a two-copy window, where S and TARGET_n agree. S is connected, because each copy meets the next and every cap vertex reaches the strip as before.

Finally S is spherical. A copy contributes three vertices, six edges and, since every face is a translated face, three faces. As $3 - 6 + 3 = 0$, $V - E + F$ is 2, as in TARGET_n . $\square$

Remark 6.2 (what is machine-checked). Three facts in the proof are checked by computation on the single certificate TARGET_n . Every face has at most five darts. No strip edge joins copies more than two apart. The rotations in the deep block are translates of one another. None is a statement about S .

6.2. What the family reaches. The floor $-(|D| - 1)$ is where deletion would consume a copy that is not deep, that is, where the two cap collars meet; each collar is two copies wide. Splicing changes the order by $3d$, so a certificate reaches one residue class modulo 3. The floors are:

$n \bmod 3$	floor	family
0	48	48, 51, 54, 57, ...
1	52	52, 55, 58, 61, ...
2	50	50, 53, 56, 59, ...

Corollary 6.3. *Conjecture 10.2 holds: there is a $(3, 4, 5)$ -alternating plane graph on n vertices for every $n \geq 20$.*

Proof. Orders below 46 are the published witnesses of [1], decoded from the corpus [7], together with the Section-8 closures of §7. Orders 37 and 38 are covered by the heuristic search of [1]; the disk surgery of §7 gives checked witnesses there as well. Orders 46 through 56, 67 through 74, 88 through 92, and 109, 110 carry the certificates of this paper. Every remaining order is reached by Theorem 6.1 from a certificate in its residue class modulo 3, with the floors above. $\square$

Remark 6.4 (the deletion direction). The floors in the table are reached by deletion. For $d < 0$ the locality argument reads as for insertion, with the four cases collapsing, except at the floor itself: there a three-copy window meets both cap collars, and not every window is a translated window. The floor orders therefore also carry their own verified certificates.

The reach is set by the hypothesis $|D| \geq 5$. Of the twenty spliceable orders, five, namely 48, 51, 53, 54 and 56, have deep blocks of only 1, 2, 2, 3 and 3 copies, so they are not eligible seeds. The eligible ones begin at 67, 68 and 69, one per residue class, so insertion reaches nothing below 70, and all ten orders 57–66 lie outside Theorem 6.1. None of them has a stored certificate. They are covered by the computation in `test_pumping_splice.py`, which builds successive splices from 90, 109 and 110 and runs each through the decision procedures. A separate proof of the deletion direction, with its own cut rule, is open.

This gives an independent construction of ([1], Theorem 8.1), which covers $n \geq 111$, and of twenty-three of the twenty-six open orders, from one certificate per residue class. Orders 46, 47 and 49 lie outside the family and

rest on their certificates alone. So the certificates and Theorem 6.1 settle *every order* $n \geq 46$ without Theorem 8.1 of [1], subject to Remark 6.4 at the five orders noted there. Orders 20 through 45 are taken from that paper.

Remark 6.5. The splice reproduces certificates it did not start from. TARGET_{110} shortened by twenty periods is TARGET_{50} up to isomorphism of rotation systems, and so is TARGET_{92} shortened by fourteen; TARGET_{90} shortened by fourteen and TARGET_{72} shortened by eight both give TARGET_{48} . So certificates found independently decompose into the same pair of caps and the same period.

7. CONJECTURE 10.3

Theorem 7.1. *There is a 3-connected alternating plane graph on n vertices for*

$$n \in \{17\} \cup [19, 56] \cup [67, 74] \cup [88, 92] \cup \{109, 110\},$$

54 orders in all, each certified by an explicit witness. (Order 17 is below the range of Conjecture 10.3; it is included because it is the smallest order at which any alternating plane graph exists, and its witness is 3-connected.)

Conjecture 10.3 asks for every $n \geq 19$. The remaining orders 57–66, 75–87, 93–108 and $n \geq 111$ are reached only by the spliced family, and we cannot prove that family 3-connected (Remark 7.3).

The witnesses come from six sources, and no order depends on more than one of them being correct.

orders	source
19	all five alternating plane graphs on 19 vertices
17, 20, 26–36, 42	published witnesses, decoded and re-verified
21, 22, 23, 25	published search seeds
21–24, 39–45	Section-8 closures, rebuilt here
37, 38	disk surgery
46–56, 67–74, 88–92, 109, 110	the certificates of §6

Order 19. Conjecture 10.2 cannot help here: [1] reports no $(3, 4, 5)$ -alternating plane graph on 18 or 19 vertices, so a witness must be a general alternating plane graph. There are exactly five on 19 vertices [6], and all five are 3-connected.

Orders 37 and 38. The heuristic search of [1] reaches every order from 20 to 42, but its Section-8 construction, of order $18a + 19b + 20c + 21d + 3$, misses these two, because 34 and 35 are not sums of 18, 19, 20, 21. So the witnesses built here are the first checkable from a stored rotation system. We obtain them by disk surgery: excise the star of a vertex, an edge or a face from a stored $(3, 4, 5)$ -alternating plane graph, let the boundary degrees float, and refill the disk.

7.1. Connectivity is not automatic. If every $(3, 4, 5)$ -alternating plane graph were 3-connected, then Corollary 6.3 would give Theorem 7.1 for all $n \geq 20$ in one step.

Proposition 7.2. *There is a $(3, 4, 5)$ -alternating plane graph on 46 vertices with a separating pair. In particular $(3, 4, 5)$ -alternating plane graphs need not be 3-connected.*

The graph is given as a rotation system in the artifact. Both verifiers accept it, its separating pair is unique, and removing that pair leaves two components of 22 vertices each. Theorem 7.1 does not depend on the proposition, since every witness is checked individually. But it means the connectivity of the spliced family needs a proof.

Remark 7.3 (the infinite tail). The natural induction shows that a freshly spliced copy is adjacent to the copies on both sides, and concludes that connectivity is preserved. What is needed is that $S(n, d) - \{u, v\}$ is connected, with the pair already removed, and that does not follow. The induction also needs the splice point to be placed clear of a pair occupying up to four consecutive copies, inside a deep block of at least five, and we have not checked this.

The witnesses at 17, 19–56, 67–74, 88–92 and 109–110 do not depend on the induction. Orders 57–66, 75–87, 93–108 and every $n \geq 111$ do. [1] states in its concluding remarks that the $(3, 4, 5)$ -alternating plane graphs of its Section 8 are 3-connected, which would cover exactly those gaps, but gives no proof. At those orders Conjecture 10.3 is open.

8. THE ASYMPTOTIC DEGREE DISTRIBUTION

The fourth problem of [1, §10] asks about the shape of a typical graph. Writing $r = v_3$, Theorem 2.5 confines v_4 to $[r - 5, \frac{3}{2}r - 5]$, so the ratio v_4/v_3 lies asymptotically in $[1, 1.5]$. Althöfer et al. ask whether there is a density function on that interval giving the asymptotic fractions of $(3, 4, 5)$ -alternating plane graphs for large n , and if so what it looks like. We cannot answer it.

All twenty-six certificates satisfy $v_5 = v_3 - 4$ and lie at the bottom of the interval. Their v_4/v_3 ranges over $[1.0000, 1.1250]$ and drifts downwards with n , reaching exactly 1 at orders 50, 56, 74, 92 and 110. This reflects the construction. The certificates are capped unrollings of one periodic strip, and Theorem 6.1 inserts copies of a single period, so every member of the family inherits that period’s degree profile.

Our methods do not reach the question. Everything proved here is an existence statement, while a density needs a measure over all graphs of order n . The class is also rigid: no two-edge switch of any of the twenty-six certificates leaves a valid $(3, 4, 5)$ -alternating plane graph, so local moves cannot walk the space.

We see three routes. An exhaustive census at small orders, with `plantri` [5] or a similar generator restricted to the degree and face-size profile, would give a first histogram of v_4/v_3 and suggest a conjecture. A structure theorem giving *every* $(3, 4, 5)$ -alternating plane graph a cap–strip–cap decomposition with a bounded interface alphabet would make the count a regular language and the density a transfer-matrix computation; we have only a construction. Failing both, a bijection with an already-counted family of plane maps would serve, and the identities of Theorem 2.5 are the natural place to look for one.

9. THE ARTIFACT

Every graph in this paper is given as a rotation system in the artifact, in a documented JSON format with a dependency-free reference reader. No file records that a graph is alternating. Degrees, faces, face sizes, bridges, connectivity and both alternation conditions are recomputed on every run by four separately written decision procedures. Three are for the $(3, 4, 5)$ class. The fourth is for the general class of Definition 2.1, which Conjecture 10.3 needs at order 19. Certificates can also be exported to `planar_code` for checking by `plantri` [5] or House of Graphs [6].

The artifact runs 1308 checks. The decision procedures use only the Python standard library, and the test suite also needs `pytest`. Each certificate is accepted by every verifier of its class, and the order-19 witnesses are rejected by the three $(3, 4, 5)$ procedures. The computations the proofs rest on are the finite hypotheses of Remark 6.2.

Data availability. The artifact, with every certificate, the four decision procedures, a drawing tool (a straight-line embedding after Tutte [4]), the export tools and the test suite, is archived at [2] under the MIT licence. The concept DOI 10.5281/zenodo.22269200 resolves to the current version; this paper describes version 1.0.4, 10.5281/zenodo.22666407. Reproduce with `make deps && make verify`. The published corpus [7] carries no licence statement, so its graphs are re-encoded in the artifact’s format, with the SHA-256 of each original kept, and the MIT licence does not cover them.

DISCLOSURE OF AI USE

Large language models were used throughout this work. They helped search for arguments and reviewed the proofs adversarially. The decision procedures and the search programs that found the certificates were written with AI assistance. The prose was drafted by AI from the author’s notes, and the author edited it. The idea of dropping (C2) (§5) came from an AI reviewer. The systems were Claude Opus 5 and Claude Fable 5.1 (Anthropic), ChatGPT at its Pro tier and Codex `gpt-5.6-terra` and `gpt-5.6-sol` (OpenAI), and DeepSeek `v4-pro` and `v4-flash`, used from February to September 2026. The author checked the mathematics and is responsible for it. No AI system is an author.

REFERENCES

- [1] I. Althöfer, J. K. Haugland, K. Scherer, F. Schneider and N. Van Cleemput, *Alternating plane graphs*, Ars Math. Contemp. **8** (2015), 337–363. doi:10.26493/1855-3974.584.09a.
- [2] H. Yang, *Alternating plane graphs: settling Conjectures 10.1 and 10.2, with witnesses for 10.3* — certificates, verifiers and test suite, version 1.0.5, Zenodo, 2026. doi:10.5281/zenodo.22269200 (concept).
- [3] B. Mohar and C. Thomassen, *Graphs on Surfaces*, Johns Hopkins University Press, Baltimore, 2001.
- [4] W. T. Tutte, *How to draw a graph*, Proc. London Math. Soc. (3) **13** (1963), 743–767. doi:10.1112/plms/s3-13.1.743.
- [5] G. Brinkmann and B. D. McKay, *Fast generation of planar graphs* (expanded edition), MATCH Commun. Math. Comput. Chem. **58** (2007), 323–357.

- [6] K. Coolsaet, S. D'hondt and J. Goedgebeur, *House of Graphs 2.0: A database of interesting graphs and more*, Discrete Appl. Math. **325** (2023), 97–107. doi:10.1016/j.dam.2022.10.013.
- [7] I. Althöfer, *Alternating plane graphs* — status page and machine-readable corpus of published witnesses, <https://althofer.de/alternating-plane-graphs.html> and <https://althofer.de/apg/apgs/>, accessed 1 September 2026.
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